Biased-noise thresholds of cubic and rhombic codes: a whole-dissertation replication audit of Pesah (2021)

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Abstract

The dissertation's principal pure-Z advantage remains only partly reconstructed. Pesah reports cubic BP–OSD thresholds of 21.7% undeformed and greater than 36% deformed; we obtain 21.908% for the former, but cannot identify a reliable asymptotic threshold for the latter. Rhombic pure-Z values are 28.4% and 6.5% in the source versus 27.407% and 6.490% here, retaining the strong disadvantage of that deformation. The preserved cubic pure-X comparison is 2.9%/7.4% versus 2.388%/7.570%. These are limited reconstructions with unknown original Monte Carlo error and production settings. A consequential point-estimate reversal occurs for cubic Sweep–Match depolarizing noise: the source deformation contrast is +0.300 percentage points, ours −0.134; our matching engine differs from the historical one. A cross-code ranking also reverses at Z bias 100, but the responsible cubic fit is unstable and does not establish a new ranking. A further 112,000 fresh trials expose size-dependent crossings; both logistic and source-family quadratic scaling fail the tested local windows. Lean proofs now classify cubic closed cochains by three cycle bits and establish minimum nontrivial cocycle weight L^2, with a finite-coordinate exactness bridge. Written analytical checks additionally recover the rhombic parameters and distances, the ideal-majority 50% limit and syndrome-consistent OSD guarantee. Actual BP–OSD asymptotic thresholds and historical-production/frontend equivalence remain unestablished. Every substantive result is accounted for, with limited or blocked outcomes retained; this is not a claim that every result was successfully reproduced.

1 Headline results and scope

Code/decoder	Noise	Source (%)		Ours (%)		$D-U$ (pp)	
		U	D	U	D	Source	Ours
Cubic BP	X	2.9	7.4	2.388	7.570	+4.5	+5.182
Cubic BP	Z	21.7	> 36	21.908	unresolved	> 14.3	unresolved
Cubic BP	Dep.	4.3	4.2	4.223	4.117	-0.1	-0.106
Cubic Sweep	X	2.9	7.1	3.031	6.816	+4.2	+3.785
Cubic Sweep	Z	14.7	22.2	14.182	19.473	+7.5	+5.291
Cubic Sweep	Dep.	4.4	4.7	4.894	4.759	+0.3	-0.134
Rhombic BP	X	1.0	1.7	1.047	1.864	+0.7	+0.817
Rhombic BP	Z	28.4	6.5	27.407	6.490	-21.9	-20.916
Rhombic BP	Dep.	1.5	1.7	1.640	1.649	+0.2	+0.009

U and D denote undeformed and deformed codes. Positive $D-U$ favors deformation. These are fitted point estimates, not established rankings. The table covers every endpoint cell of Tables 3.1–3.2; the

deformed cubic Z source is an unresolved lower-bound statement, not an exact 36% measurement. Ours-minus-source deviations are a different quantity, reported later. Fits with optimizer or model diagnostics are retained as outputs, not promoted to reliable thresholds.

Arthur Pesah's *Tailoring 3D topological codes and decoders to biased noise* is a University College London *Master of Research dissertation*, submitted 1 September 2021 under Dan Browne, not a doctoral thesis (Pesah, 2021). We read all 66 pages. The initial milestone selected only cubic pure-X BP–OSD. The present objective covers every substantive dissertation result: endpoint thresholds, finite-bias comparisons, total/point/loop curves, code constructions and analytical reductions, the OSD optimization and the graphical interface. The original pure-X paper, code and results remain preserved. Background exposition and explicitly proposed future experiments are identified separately in `coverage.json`; none is silently counted as a reproduced original experiment.

The scientific question is whether local Hadamard deformation improves noise tolerance without changing physical-qubit count. The answer depends on code and channel. Cubic pure-X improves in the reconstruction; rhombic pure-Z worsens strongly. The larger cubic pure-Z claim remains unresolved. Resource equality, finite stabilizer checks, or agreement in one channel cannot substitute for that missing performance result.

2 Experimental reconstruction

We run periodic sizes $L = 4, 6, 8, 10$, both deformation arms, perfect syndrome measurements and independent physical-qubit errors. Categorical I/X/Y/Z draws preserve the simultaneous X and Z components of Y. For bias η along one Pauli axis, its probability is $p\eta/(\eta + 1)$ and each other axis receives $p/[2(\eta + 1)]$. Depolarizing noise is $\eta = 1/2$; the pure channels are the limiting cases. Hadamard deformation swaps X and Z on the selected axis.

The primary decoder reconstructs the historical ten-iteration BP and order-zero OSD implementation in Huang et al. (2021); its algorithmic context is Roffe et al. (2020). The source does not identify production seeds, counts, iteration settings, package versions, tie handling or fitting windows. Current NumPy quicksort, the historical prior regularizer and independent PCG64/MT19937 trials are explicit supplied assumptions. Full decoder equivalence to the undisclosed source production run is not established.

An original C++ GF(2) elimination routine accelerates OSD while preserving reliability ordering, pivot selection and zero free variables. It matched the preserved Python routine in 196 correction cases through $L = 10$. The generalized cubic decoder matched 480 historical dense-reference cases; the rhombic implementation matched 320, after an explicit global Pauli conversion. These checks share graphs, priors and the stated algorithm. They independently check arithmetic/correction implementations, not an entire independent scientific replication.

The rhombic convention is consequential: the thesis prose uses cube Z checks and triangle X checks; the available historical code globally interchanges them. We use the prose convention and verify the exact global-Hadamard mapping. A nearby figure caption also conflicts with the prose. Table 3.2 calls the lattice cubic although its surrounding section and values describe rhombic. Figure 3.6 labels the undeformed pure-Z endpoint 21.0 percent versus Table 3.1's 21.7 percent; Figure 3.9 labels deformed depolarizing 1.6 percent versus Table 3.2's 1.7 percent. We retain table values for endpoint targets and record the figure discrepancies, rather than silently selecting whichever agrees best. This choice explains our channel labeling but does not prove which code revision generated the published plots.

Historical Sweep code is executed externally with PyMatching 2.3.1. The original 0.2.4 build required a LEMON archive whose host denied access. We did not retrieve it through another route. The modern sparse-blossom implementation (Higgott and Gidney, 2025) is a disclosed engine departure. Eighty finite

matching checks agree with independent exhaustive pairing of up to six defects, using shared shortest-path distances and graphs; they do not prove historical tie equivalence. All nonzero residual syndromes count as failures: 2,696 of 19,200 Sweep trials retain a residual syndrome. These are retained decoder failures, not discarded trials; all BP production corrections satisfy their syndromes.

The pure-Z grid contains 8,800 fresh trials and took approximately 2,478 seconds. The 28 BP families contain 179,200 trials; three Sweep families contain 19,200. Three wider diagnostic grids add 38,400. A further 5,600 shared-error diagnostic trials compare ten versus 100 BP iterations and stable versus quicksort ties (16,800 decoder executions). The full source-vertex curve run contains 226,800 trials; these production and wider-grid runs total 472,400. Individual production processes are capped at 2 GiB address space and 2,400 CPU seconds, with bounded worker concurrency. These counts exclude the preserved 16,000-trial pure-X milestone and small validation/pilot runs. Exact measured times, seeds and completion flags reside with each output. All production files passed completion, row-cardinality and count-integrity checks. Across separately reported families, 406 generator/seed keys are reused; those families are not pooled as independent evidence. The six cross-code contrasts use disjoint simulation seeds. Bonferroni requires no cross-test independence.

3 Numerical agreement and uncertainty

The source fits a quadratic finite-size form $y = A + Bx + Cx^2$, $x = (p - p_{\text{th}})L^{1/\nu}$, by unweighted least squares (equations 3.1–3.2; see Tuckett et al., 2020). We retain the initial bounded fit and separately report an exploratory normalized-x reparameterization with relaxed coefficient/exponent bounds. Fits drop individual probability endpoints and extreme sizes as diagnostics. The normalized fit is not silently substituted for the original estimator.

Each two-arm family has 400 conditional binomial bootstrap draws. Bounds, nonphysical predictions and grid extrapolation prevent robust inference in several families. The expanded source-comparison family has 82 quantities: 24 BP families times three quantities, three Sweep families times three, and the undeformed cubic Z threshold. The preserved pure-X three-comparison family stays separate. We save unadjusted and Bonferroni-adjusted conditional bands; normal bands based on bootstrap standard errors are approximations, explicitly withheld as inference when diagnostics fail. Six cross-code contrasts form a separate family. These adjustments neither supply missing source uncertainty nor control finite-size/model bias.

Family	Arm	Ours − source (pp)	Conditional band (pp)	Rounding
cubic dep.	U	-0.077	[-0.66, +0.51]	close
cubic dep.	D	-0.083	[-0.57, +0.40]	close
rhombic dep.	U	+0.140	[-0.31, +0.59]	close
rhombic dep.	D	-0.051	[-0.38, +0.28]	close
rhombic X	U	+0.047	[-0.11, +0.20]	match
rhombic X	D	+0.164	[-0.77, +1.10]	mismatch
rhombic Z	U	-0.993	[-4.02, +2.03]	mismatch
rhombic Z	D	-0.010	[-0.92, +0.90]	match
sweep X	U	+0.131	[-0.75, +1.01]	close
sweep X	D	-0.284	[-1.61, +1.04]	mismatch
sweep Z	U	-0.518	[-1.63, +0.59]	mismatch

Family	Arm	Ours − source (pp)	Conditional band (pp)	Rounding
sweep Z	D	-2.727	[-4.75, -0.70]	mismatch
sweep dep.	U	+0.494	[-0.18, +1.17]	mismatch
sweep dep.	D	+0.059	[-0.51, +0.62]	close
cubic Z	U	+0.208	[-3.28, +3.70]	mismatch

The displayed bands concern ours-minus-source deviations, in percentage points. They exclude unknown source Monte Carlo error. Table rounding uses half-up 0.1 pp precision: a match has identical rounding, and close is one last-place unit. Rounding remains distinct from uncertainty overlap and limited reconstruction scope. Figure coordinates have a conservative 0.1 pp extraction allowance, not an asserted original precision or trial error. Difference contrasts accumulate both input allowances.

Undeformed cubic pure-Z has bootstrap percentile interval 20.884–24.996%, with four boundary fits among 400 draws; deleting an extreme size moves its estimate to 20.745–23.082%. The deformed curve saturates and its sector fits hit boundaries. A numerical polynomial output around 42% is therefore not reported as a resolved threshold. The source itself describes the 36–50% crossing as uncertain.

Rhombic pure-Z yields a signed deformation contrast −20.916 pp versus source −21.9 pp. Its bootstrap fits have no optimizer-bound hits in this grid, but configuration and finite-size limits remain. The small rhombic depolarizing fitted advantage, +0.009 pp versus source +0.2 pp, does not establish a meaningful ranking. Sweep depolarizing changes from source +0.300 pp to ours −0.134 pp; seven bootstrap fits hit bounds, and modern matching further limits the comparison.

4 Finite bias and cross-code comparisons

Code/axis	η	Source (%)		Ours (%)		$D - U$: source/ours (pp)
		U	D	U	D	
Cubic X	1	3.52	4.12	3.70	4.15	+0.60/+0.45
Cubic X	3	3.33	4.70	3.11	4.73	+1.37/+1.61
Cubic X	10	3.04	5.68	2.96	5.61	+2.64/+2.65
Cubic X	30	2.86	6.75	2.97	6.27	+3.89/+3.30
Cubic X	100	2.89	6.81	2.93	6.94	+3.92/+4.02
Cubic Z	1	5.56	5.08	5.57	4.94	-0.48/-0.63
Cubic Z	3	12.12	7.25	11.66	8.50	-4.86/-3.16
Cubic Z	10	22.35	12.44	22.51	13.19	-9.92/-9.31
Cubic Z	30	21.74	16.00	21.53	19.71	-5.74/-1.82
Cubic Z	100	21.71	23.75	21.03	37.83	+2.04/+16.80
Rhombic X	1	1.27	1.44	1.45	1.64	+0.17/+0.20
Rhombic X	3	1.16	1.50	1.22	1.63	+0.34/+0.41
Rhombic X	10	1.09	1.47	1.21	1.70	+0.37/+0.49
Rhombic X	30	1.07	1.59	1.04	1.80	+0.52/+0.76
Rhombic X	100	1.00	1.63	1.67	1.87	+0.63/+0.20

Code/axis	η	Source (%)		Ours (%)		$D-U$: source/ours (pp)
		U	D	U	D	
Rhombic Z	1	2.09	1.80	2.15	1.84	-0.29/-0.31
Rhombic Z	3	4.00	2.47	4.59	2.59	-1.53/-1.99
Rhombic Z	10	11.14	3.90	11.95	3.86	-7.25/-8.08
Rhombic Z	30	28.94	5.11	28.52	5.53	-23.82/-23.00
Rhombic Z	100	28.00	6.00	29.93	5.67	-22.00/-24.26

Figure-derived source values above retain two decimal places only for readability; their extraction allowance is 0.1 pp. These are all plotted finite-bias conditions, with both arms and all four lattice sizes. The source methods list $\eta = 5$ but vector plot coordinates put the corresponding point at 3. We run both; only 3 is compared with that plotted coordinate. Separate eta-5 results remain an alternative-method reconstruction without an invented source target.

Three wider independent grids were chosen after unstable fits. They remain exploratory: cubic Z eta-5 undeformed gives 18.905% but nonphysical/nonmonotone polynomial predictions; cubic Z eta-100 deformed reaches the 49.99% optimization bound outside the sampled grid; rhombic X eta-100 undeformed gives 0.725% with some negative fitted probabilities. None resolves the affected original estimate simply by adding shots over a broader range.

The source ranks undeformed rhombic highest at Z bias 30–100. Five of six reconstructed pairwise point contrasts retain that direction. At eta-100, undeformed rhombic minus deformed cubic is -7.898 pp, reversing the source point ordering. The cubic fit is not identified reliably, including in the wider-grid check; this is evidence that the ranking is not reproduced robustly, not evidence for a new asymptotic winner. No source ranking is statistically established without its uncertainty.

5 Fresh diagnostics for unstable finite-size fits

We froze new grids using previous size-crossing brackets, without source thresholds as inputs, and ran 112,000 fresh trials at all four sizes. The two local families use 1,000 trials per generator and condition; the broad sector study uses 500. The 144 generator/seed pairs do not overlap the earlier expanded trials. Largest-size pilots used 60 trials each and cost 3.170, 0.364 and 2.460 seconds; two-worker production took 722.614 seconds, with peak resident memory below 70 MiB per process. Every BP correction was syndrome-valid. All earlier primary and wider-grid results remain unchanged.

Family	Local quadratic (%)	Logistic (%)	Deviance/16 df
Cubic Z, $\eta = 5$, U	20.5414	20.5725	137.219
Rhombic X, $\eta = 100$, U	1.05866	1.08223	56.699

These are diagnostic fitted outputs, not accepted thresholds. The quadratic curves are physical and monotone on these frozen grids, unlike the earlier broad fits, but that alone does not establish scaling. The independently fitted bounded model is $\mathrm{logit}(P) = a + b[(p-t)/\,\mathrm{median}(p)](L/6)^{1/\nu}$. Its approximate goodness-of-fit probabilities are 2.52×10^{-21} and 1.86×10^{-6}: both reject this model. Its profile intervals therefore are not reliable threshold intervals. A separate frozen parametric-bootstrap test now assesses the quadratic estimator itself. Counts are drawn from each fitted quadratic curve, and the identical

five-start unweighted estimator is refitted. Observed binomial deviances are 138.190 and 41.372; none of 2,000 refits per family exceeds them. The add-one tail estimate is 0.000500, with Monte Carlo 95% upper bound 0.001843 for each. All 4,000 refits pass the declared validity checks; failed or nonphysical fits would have bounded, not disappeared from, the tail calculation. This rejects both quadratic models at the two-model Bonferroni level and the stated 0.005 sensitivity. It is an exploratory plug-in diagnostic, not an exact finite-sample test or threshold interval. Production took 53.6 seconds after excluded pilots; no physical-error trials or windows changed. We stop fitting these grids for an established threshold; a justified alternative model and prospective size/grid design are required. Omitting the smallest or largest size is preserved as sensitivity, not used to select a favorable result. There is no unique plotted eta-5 source value because of the methods/figure discrepancy.

For cubic eta-5, the $L = 4/6$, $4/8$ and $4/10$ interpolated crossings are 23.0704%, 21.7571% and 20.3394%; the remaining pairs fail the predeclared uniqueness/stability guard. Corresponding rhombic crossings are 1.16486%, 1.12861% and 1.07830%, with the other pairs likewise failing. A 12-member Bonferroni bootstrap family covers finite-grid interpolation conditionally on a unique root; it is not asymptotic threshold coverage. Missing, multiple or oppositely oriented crossings are retained, not fitted away.

Deformed cubic Z at eta-100 illustrates separate sector changes. At $L = 10$, total/point/loop failure is 0.494/0.494/0 at $p = .20$, 0.521/0.513/0.018 at .28, and 0.887/0.513/0.764 at .40. Eleven of 72 adjacent-size/sector signed differences exclude zero under conservative simultaneous exact-binomial intervals. All comparisons remain in the archive. No single global threshold is forced onto these curves, and the descriptive separation is not a causal or asymptotic proof. Point and loop failures share trials. Separate scalar likelihood and raw-count aggregation checks reproduce the analysis; they share the observations and model definitions, not an independent physical simulation.

6 Full curve and analytical coverage

The four pure-channel/code combinations cover total, point and loop failure for both arms and every size. The source supplies 6,804 factual vector coordinates across 96 curves. The reconstruction samples their horizontally displayed probability vertices, snapping calibrated coordinates to the source's 0.005 grid within extraction tolerance. Below-axis source values are treated as bounds rather than exact zero observations. The final comparison uses a declared 6,804-member Bonferroni Clopper–Pearson family for our binomial counts plus a 0.005 vertical extraction allowance; source trial counts remain unknown. All 6,804 source-coordinate intervals overlap our simultaneous bands, with zero invalid BP corrections in these 226,800 trials. This is a low-resolution compatibility check: only 100 fresh trials per condition and a large multiplicity family produce wide bands. It is not numerical equivalence, a recovered source trial archive, or evidence that every plotted value was exactly reproduced. The source has 4,322 visible vertices; the remaining below-axis coordinates are treated conservatively as probability intervals [0,0.015].

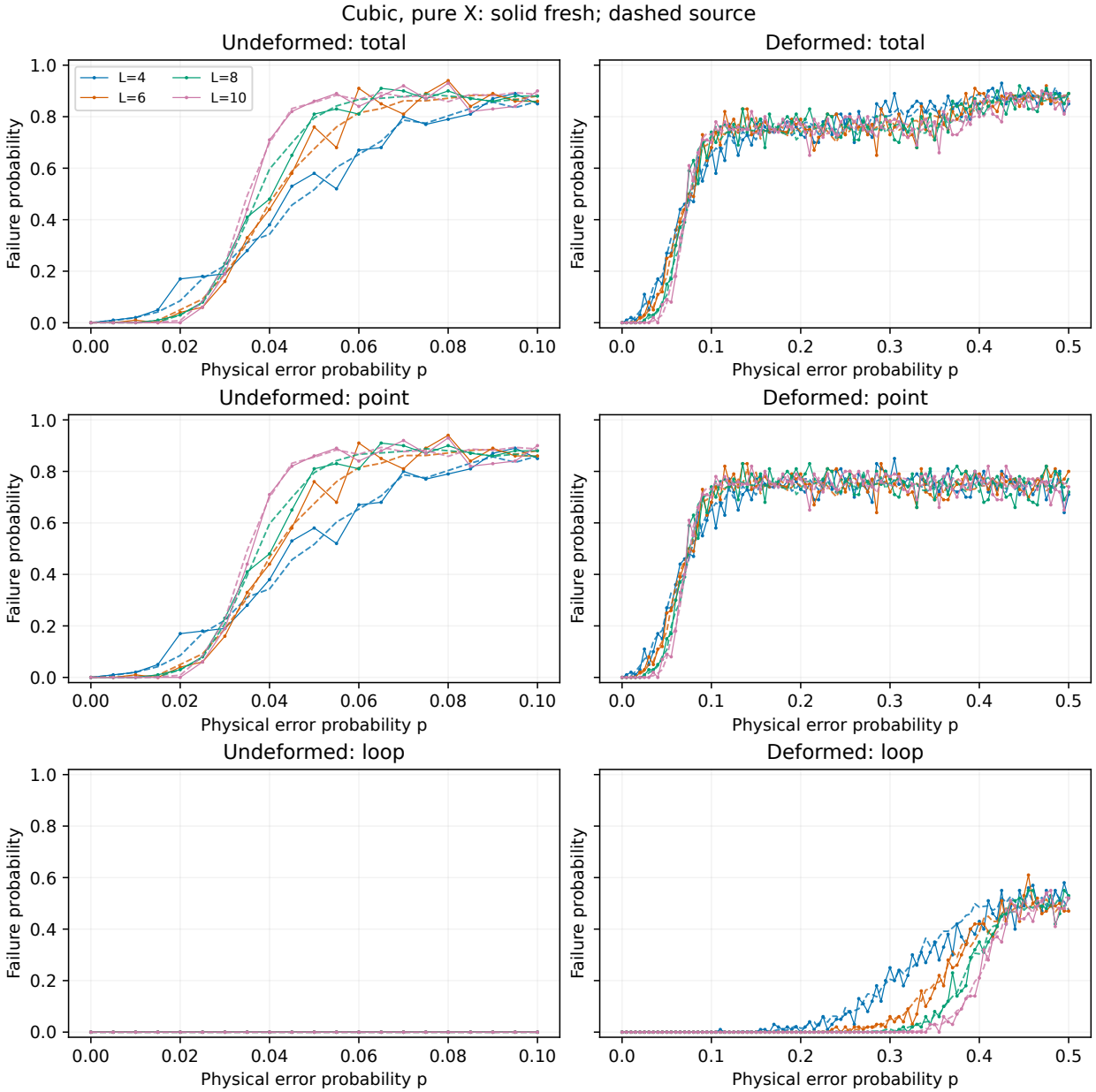


Figure 1: Cubic pure X: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

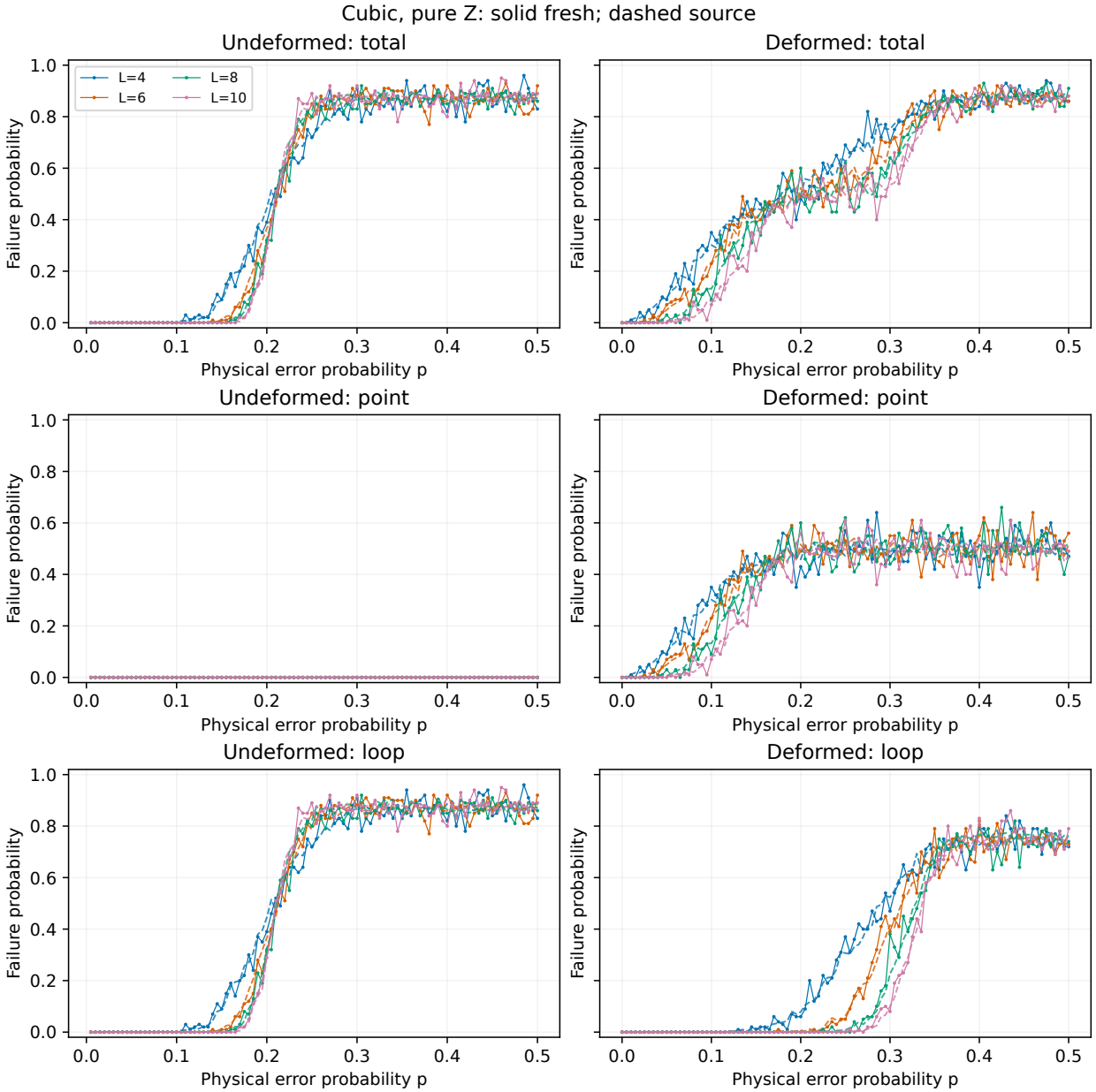


Figure 2: Cubic pure Z: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

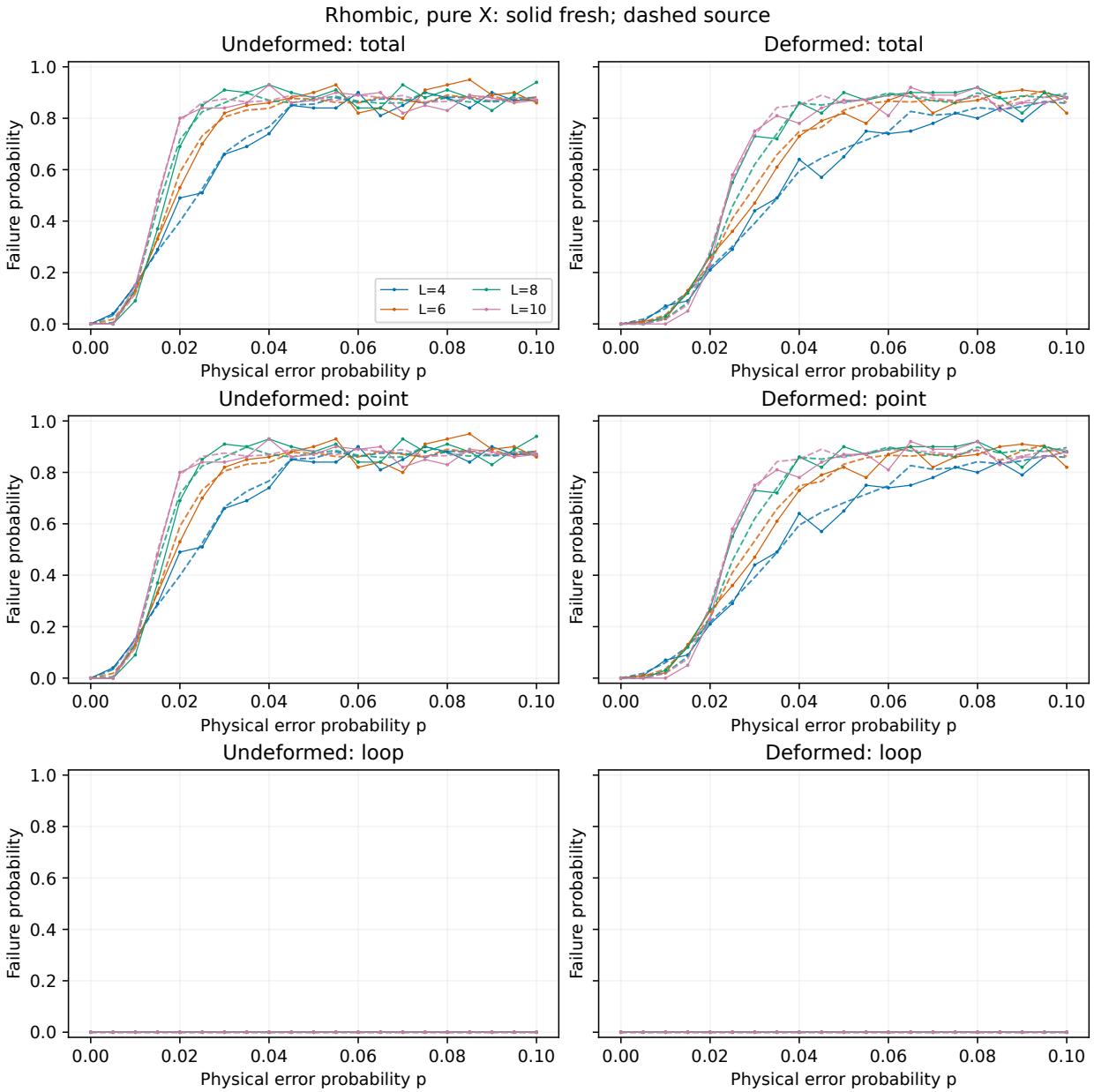


Figure 3: Rhombic pure X: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

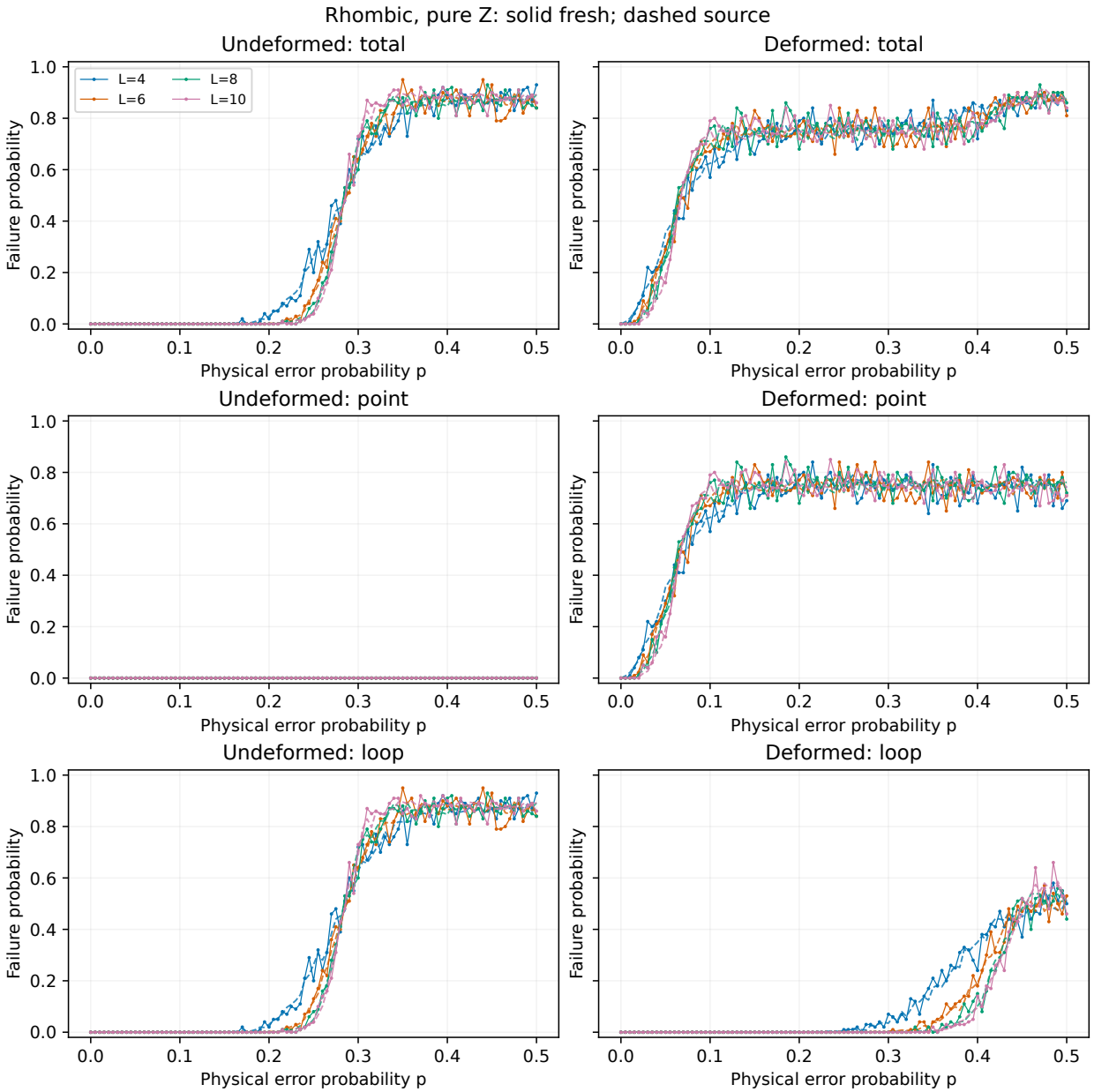


Figure 4: Rhombic pure Z: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

Finite geometry checks establish commutation, three logical qubits, matched logical operators and restricted-axis nullities at $L = 2, 4, 6, 8, 10$. Computational distance certificates verify matching lower and upper bounds at these sizes: cubic $d_X = L$, $d_Z = L^2$, and rhombic $d_X = L$, $d_Z = 2L^2$ under our convention. The lower-bound check uses disjoint homologous representatives and exact row-space membership. These are finite computational certificates, not Lean proofs or an arbitrary-L theorem.

For the ideal repetition-chain and plane reductions, a separate analytical audit evaluates finite binomial majority probabilities and parity composition, with explicit independent-component and random-tie assumptions. The 50% ideal-sector limit is not the threshold of the actual BP–OSD decoder, nor does it establish the total-code threshold. The rhombic pure-X loop claim is checked through finite restricted-axis nullities. The follow-up now proves general binary Hadamard preservation and restricted difference-kernel formulas in Lean 4.22.0; the exact source bridges and remaining gaps are in Appendix B. The latest continuation additionally proves the cubic coordinate cohomology classification and cocycle distance described below. The next section supplies written quantum, rhombic and probability arguments, separately from Lean coverage.

6.1 General cubic cohomology and distance

The source states $k = 3$, $d_X = L$ and $d_Z = L^2$ for the periodic cubic construction (Section 2.4.1, pp.28–29). Our new general Lean result classifies closed periodic binary cochains by exactly three Boolean cycle labels, modulo vertex gradients, and proves minimum nontrivial cocycle weight L^2. A finite-coordinate bridge connects zero holonomy to the actual vertex-gradient formula, including wraparound. This advances the Z-sector argument beyond the earlier finite certificates; it is not a claim that all quantum-code semantics have been formalized.

Source claim	Reconstructed consequence	Evidence class
Three logical pairs	Three independent cycle labels	General coordinate Lean classification
Z distance L^2	Minimum nonzero cocycle weight L^2	General Lean lower bound and witness
X distance L	Dual cycle/slab argument	Written proof; formal duality pending
$k = 3$	Vertex/face ranks $L^3 - 1$, $2L^3 - 2$	Written rank consequence; formal bridge pending

A separate consumer checks all eight labels at odd and even sizes $L = 2, 3, 4, 5, 6$, using two input potentials: 80 fields and 21,120 gradient entries agree with the Python matrices. It checks nonzero period jumps rather than incorrectly assuming every closed field has a periodic potential. These constructed examples share inputs and geometric conventions; they validate implementation connections, not the general theorem by sampling. Appendix C gives the constructive proof and exact remaining boundaries.

7 Written verification of the remaining analytical claims

The remaining code-parameter and ideal-decoder statements admit written verification, while the actual BP–OSD asymptotic endpoint does not follow from them. Table 3 separates these outcomes. The complete arguments and assumptions are in `analytical-completion.md`; these are written proofs, distinct from the kernel-checked statements above. Standard torus homology, finite binary rank-nullity and Pauli trace identities are inherited background, not new formalization targets. The runnable revision is `README.md` at `60ccabae6c835f90a627341099fec1215856acb2`.

Source statement	Verified result	Evidence and boundary
Rhombic parameters, pp30–32	$n = 3L^3$, $k = 3$, $d_X = L$, $d_Z = 2L^2$ for even $L \geq 2$	Written FCC cellulation and disjoint-cycle proof; independent matrices at five sizes. Prose Pauli convention.
Cubic reductions, p44	2D point layers; ideal chain/plane majority endpoint 50%	Existing kernel proofs plus written probability argument; 48 exact-rational comparisons. Not the actual BP threshold.
Rhombic loop, pp55–56	Unique recovery on first-axis support	Restricted formula proof. Four outside-grid stress failures qualify unrestricted implementation wording.
OSD, pp34–40	Correct syndrome when $s \in \operatorname{im} H$	Written elimination invariant; 20,748 exhaustive cases across three solvers. No optimality or BP convergence claim.
Hadamard, pp25,44	n, k, d preserved	Binary Lean transport plus written unitary/projector bridge. Mixed-code sector distances not asserted invariant.

Table 3: Analytical statements from Pesah (2021), with our verification and its limits. Written arguments and finite checks are not relabeled as Lean proofs.

7.1 Rhombic homology and distance

The odd-parity cubes define vertices of a face-centred cubic complex. Each physical edge touches exactly two such cubes and becomes their connecting edge. The four odd cubes around an original vertex form a tetrahedron: its six edges are the six incident physical edges, and its four faces are the source triangle checks. Cube and triangle incidence therefore commute. Subdivide each unit cube into its odd-corner tetrahedron and four even-corner pieces. The latter join into octahedra. This fills space; even periodic identification gives a three-torus, including small quotients with identified edge endpoints.

There are $L^3/2$ vertices and $3L^3$ edges. Connectivity gives cube rank $L^3/2 - 1$; three-dimensional first homology gives triangle rank $5L^3/2 - 2$, hence $k = 3$. A nontrivial cycle has a trail with displacement at least L in one coordinate, while each edge changes that coordinate by at most one. A length-L zigzag attains the resulting X-distance. For each winding direction there are $2L^2$ edge-disjoint homologous zigzags, one for each transverse position and each of two physical edge orientations. A nontrivial cocycle intersects all of them oddly, so its weight is at least $2L^2$. A transverse cut with both orientations attains this Z-distance. The source's prose and figure/historical Pauli labels conflict; these formulas follow the prose convention.

`check_rhombic_cellulation.py` reconstructs endpoints and tetrahedron faces independently of the source corner-triple implementation. At $L = 2, 4, 6, 8, 10$, every cube matrix and triangle multiset matches, and all counted zigzags have connected endpoints, correct winding and disjoint supports. The check takes 0.289 seconds and under 84 MiB RSS. Its shared periodic-coordinate convention limits independence; finite cases support the mapping, while the general result uses the written cellulation proof.

7.2 Ideal majority and actual decoding

For a component of m independent Bernoulli(p) errors, let $q_m = \Pr[B > m/2] + \frac{1}{2}\Pr[B = m/2]$, where $B \sim \mathrm{Bin}(m, p)$ and ties use independent fair coins. Parallel components represent the same logical class,

so the failure probability for N components is

$$P_{\text{fail}} = \frac{1-(1-2q_m)^N}{2}.$$

The restricted cubic sectors have $(m, N) = (L, L^2)$ for chains and (L^2, L) for planes. For fixed $p < 1/2$, $q_m \le \exp[-2m(1/2-p)^2]$ and $P_{\text{fail}} \le Nq_m \to 0$. One direct derivation uses exponential Markov inequality: for $0 < p < 1/2$, choose $t = \log[(1-p)/p]$ to bound $\Pr[B \ge m/2]$ by $[2\sqrt{p(1-p)}]^m \le \exp[-2m(1/2-p)^2]$; $p = 0$ is immediate. At $p = 1/2$ the logical parity is uniform. This proves the ideal-majority endpoint on $0 \le p \le 1/2$, not that ten-iteration BP–OSD realizes it. The remaining point equations at fixed first coordinate are exactly square-lattice vertex incidence; `check_cubic_layers.py` independently verifies the finite mapping.

`check_majority_exact.py` independently evaluates Bernoulli polynomial convolution and parity recursion using rational arithmetic. All 48 saved predictions at four sizes and six probabilities agree within 1.43×10^{-15}; the 2×10^{-14} check tolerance concerns floating-point evaluation, not sampling uncertainty. The argument requires independent errors and independent tie coins. Further finite-size fits cannot turn this ideal-decoder proof into a BP threshold theorem.

In the deformed rhombic loop sector, each first-axis error is exposed by a triangle, proving unique recovery restricted to that support. An unrestricted regularized decoder can nevertheless leave it. At $L = 2$, exhaustive tests of 256 patterns at prior 10^{-12} find four valid-syndrome logical failures; the external historical dense implementation reproduces all four sparse-decoder corrections. At priors $.001, .1, .49, .5$ every pattern succeeds. `check_loop_reference.py` and `rhombic-loop-exhaustive.json` retain the witnesses. The source wording is preserved in `rhombic_loop_argument.png` and `rhombic_loop_continuation.png`. These stress cases lie outside the dissertation's sizes and probability grids: they qualify a universal implementation guarantee, not its reported zero-loop-error observations or the restricted uniqueness theorem.

7.3 OSD and the quantum bridge

Column permutation and invertible row operations preserve the system $Hc = s$. If $s \in \operatorname{im} H$, the zero rows after elimination have zero right-hand side. Setting free variables to zero and pivot variables to their reduced right-hand side solves every equation; undoing the permutation yields a valid correction. The native solver's skipped prefix words are zero by its active-row invariant. The source/Python solvers omit an inconsistency guard, so the consistent-syndrome premise is essential; all production syndromes here satisfy it.

An independent column-span enumeration in `check_osd_exhaustive.py` checks all 682 binary matrices of shapes 1–3 by 1–3, every consistent syndrome, all strict reliability orders and a tied order. All 20,748 cases match Python, native and historical solvers: 62,244 correction comparisons. In 455 inconsistent examples native rejects while source/Python return invalid corrections. The complete check takes 3.34 seconds and under 48 MiB RSS. Shared matrices and tie ordering are disclosed. This verifies consistency, not optimality or convergence of approximate BP.

For r independent commuting Hermitian Pauli checks generating no $-I$, the common-eigenspace projector is $2^{-r}\prod_i(I+g_i)$. Only its identity term has nonzero trace, giving dimension 2^{n-r}. Independent CSS bases satisfy the no-$-I$ premise because a scalar product requires both binary supports to vanish. Tensor-product Hadamard conjugation is unitary and preserves Pauli support, including the irrelevant sign on Y. This supplies the written Hilbert-space bridge to the earlier binary transport proof; it does not assert unchanged CSS sector distances after mixing.

7.4 What remains unresolved

All seven analytical families now have an explicit verified statement and boundary. General cubic rank/X-distance, rhombic homology/distances and quantum interpretation are written checks; their absent full Lean formalizations are not missing original experiments. The actual BP–OSD 50% asymptotic expectation remains unestablished. The frozen numerical diagnostics reject the tested local scaling models; changing windows to obtain a preferred threshold is not justified. Original production counts/configuration, benchmark inputs and the denied frontend dependency remain the concrete missing inputs. Circuit-noise and other Chapter 4 proposals are future work, not experiments omitted from this audit. The 87-item register accounts for these distinctions without claiming all results were reproduced successfully.

8 Software contributions and attempted blockers

The source claims a tenfold OSD speed improvement from reusing elimination rather than recomputing ranks. Our reconstructed repeated-rank baseline and reused-elimination implementation agree on 15 small cases and show speedups roughly 12–145 times. The original benchmark inputs, baseline implementation and hardware are absent; concurrent workload also affects our timings. This supports the mechanism, not an exact tenfold benchmark reproduction. The native optimization's separate speedup is not credited as a reproduction of that source claim.

On shared diagnostic errors, total failures are 1,375 at ten iterations, 1,476 at 100, and 1,353 with stable sorting, out of 5,600 each. These heterogeneous exploratory counts demonstrate setting sensitivity; they are not pooled threshold estimates. No invalid correction occurred in these checks.

The historical Flask backend passes 28 response checks: four code/size geometry checks and 24 zero-syndrome BP decoding requests. Browser loading of the original frontend encounters a denied external Three.js module host and produces no canvas. Frontend interaction/rendering remains blocked; no alternate CDN was used. Backend success is not full GUI reproduction. Claimed usefulness for visualization is qualitative and lacks a source evaluation protocol.

Chapter 4 proposes measurement/circuit noise, coprime dimensions, boundaries, higher-order OSD, a statistical-physics model and future resource comparisons. They are future work, not dissertation experiments that we failed to run. We do not invent circuit-level thresholds or Y-bias results merely because the setup mentions general Pauli channels.

9 Availability, compliance and next steps

The full source and external code remain outside this repository. Limited attributed evidence crops, vector factual coordinates and original reconstruction artifacts are governed by the repository's licensing notices. Source evidence: `table3_1.png`, `scaling_equations.png`, `table3_2.png`, `cubic_bias_figure.png`, `rhombic_bias_figure.png`, `methods_grid.png`, `manifest.json`, `expansion-manifest.json`. The following rows are copied from `expanded-availability.md` and `expanded-compliance.md`. The original milestone ledgers remain separate.

Item	Result and limit
Thesis and production archive	Complete 66-page author-hosted MRes source obtained at b4f401e0c12b44be4c69318e0015f70d8a3d3f98. All 47 returned author repositories and historical collaboration code searched; no pinned production trials/settings archive identified. Earlier search ledger retained; absence is not exhaustive.
Historical decoder source	PanQEC revision 686560e5d03c2e5d31f6c6260b03756af03a7e93 obtained externally. Dense BP and Sweep routines executed without vendoring. Exact dissertation production identity unknown.
Historical matching dependency	PyMatching0.2.4 could not be built: original setup fetched a LEMON archive from lemon.cs.elte.hu and received HTTP403. No alternate route or retry. PyMatching2.3.1 is a labeled reconstruction, not historical equivalence.
Graphical interface	Historical Flask backend tested. Original frontend requires cdn.skypack.dev/three@v0.130.1; CONNECT403 prevented module access and canvas rendering. No alternate CDN or host retry.
Unspecified research inputs	Original trial counts, seeds, BP settings, exact package pins, fitting windows and OSD benchmark workload/hardware were not found. Source uncertainty remains unknown. No author contact.
Permitted evidence	Eight attributed 600-dpi crops and factual vector coordinates retained with provenance; complete thesis and third-party source remain external. THIRD_PARTY.md and LICENSING.md apply.

Item	Result and limit
Objective and source status	Whole-dissertation objective, not selected-result completion. 87-item register includes numerical, analytical and software claims plus explicit background/future exclusions. Source is an MRes dissertation, not PhD; this selection departed from initial doctoral screening without asserting user-specific selection.
Skill and template	Replication skill and approved-preamble asset at revision 1e69e2c0e6fa2f42d0879d48936561df02659add; approved Kupzyk layout retained. Quantum BP/OSD and matching engines replace inapplicable SEM engines with recorded preflight. No reusable skill edits.
All numerical families	All source endpoint cells, finite plotted biases, eta5 method alternative and all pure-channel sector/arm/size curves attempted. Expanded production and wider grids: 472400 trials. Original pure-X milestone preserved. Unstable fits remain unresolved, not counted as successful thresholds.
Independent validation	480 cubic and 320 rhombic historical dense correction matches; 196 native/Python OSD cases through L10; 80 modern matching/exhaustive-pairing cases. Shared inputs, graphs, priors and some software explicitly limit independence. Finite code/distance checks do not establish general theorems.

Item	Result and limit
Resources and supplied defaults	Pure-Z pilot approximately 8.43/9.10 seconds per ten L10 trials; production 2478 seconds. Native optimization piloted before remaining families. Per-process 2400 CPU seconds and 2GiB address space, bounded workers and deterministic checkpoints. Ten BP iterations, order-zero OSD and current tie behavior are supplied assumptions.
Signed contrasts and precision	D minus U in pp favors deformation when positive. Separate ours-minus-source deviations. Sweep-depolarizing and eta100 cross-code point reversals disclosed with instability/engine limits. Printed table rounding, figure extraction, source uncertainty and reconstruction scope remain separate.
Uncertainty families	82 expanded threshold/source quantities, six cross-code contrasts and 6804 curve-coordinate checks, each alpha0.05 Bonferroni; preserved pure-X family separate. Conditional bootstrap-normal bands flagged when fits fail diagnostics. Curve counts use exact binomial intervals. Missing source Monte Carlo and systematic errors are not repaired by multiplicity correction.
Sampling and failures	Distinct seeds within each run; 406 generator/seed keys overlap separately reported families, which are not pooled independently. Cross-code comparison seeds disjoint. All 2696 residual-syndrome Sweep trials count as failures; no invalid BP production correction.
Preservation and remaining work	Original pure-X code, analysis, bibliography and PDF preserved. Normalized and wider-grid diagnostics remain separate. Written analytical verification now complements Lean and finite checks. Actual BP asymptotic thresholds, historical runtime/production equivalence and frontend rendering remain unestablished.
Manuscript and delivery	Expanded main manuscript leads with headline numbers and limits, contains readable full-condition tables, source evidence and copied ledgers. Rendered PDF/scientific review and remote hash verification are separate acceptance gates; a successful build alone does not establish delivery.
Fresh follow-up	112000 additional trials, zero invalid corrections, 722.614 seconds with two workers. Frozen source-independent grids; 12 crossing and 72 sector-contrast families. Logistic models rejected; no replacement threshold claim. Lean4.22 binary algebra and restricted kernels checked; remaining written lattice/semantic/probability checks appear in the final analytical section.
Proof and adequacy continuation	Explicit user permission resolved the public-claim prerequisite. General cubic incidence and restricted kernels checked in Lean; 609792 cross-language matrix entries match. Two fixed quadratic models fail a 4000-refit bootstrap diagnostic; no windows or physical trials changed. Subsequent written homology/counting/probability checks are distinguished from Lean coverage. Required branch authorization should be identified upfront; proposed guidance recorded without reusable skill edits.

Item	Result and limit
Cubic homology continuation	General periodic cochains classified by three Boolean cycle labels; exact nontrivial cocycle minimum L^2 and finite-coordinate exactness bridge proved in Lean. 80 independent consumer fields and 21120 gradient entries checked. Standard Classical.choice now appears in existence axioms. Rank/X-distance written consequences remain separate; subsequent written arguments close analytical bridges without claiming full Lean coverage.
Final analytical verification	All seven analytical families assessed with source-specific stopping rules. Written rhombic and quantum proofs, exact-rational majority checks and exhaustive OSD validation supplement prior Lean results. Actual BP asymptotic claims remain unresolved. Four outside-grid loop witnesses qualify unrestricted implementation wording. No source production inputs or frontend access were invented.

The register now records a result or explicit limit for every substantive item. Actual BP asymptotic endpoints and reliable scaling for the rejected local models remain unestablished. Exact historical production reproduction requires the missing job inputs, trial archive and environment; the matching and GUI source-access blockers remain. These are unresolved scientific or access outcomes, not successes concealed by the analytical checks.

References

Oscar Higgott and Craig Gidney. Sparse blossom: correcting a million errors per core second with minimum-weight matching. *Quantum*, 9:1600, 2025. doi: 10.22331/q-2025-01-20-1600. URL https://quantum-journal.org/papers/q-2025-01-20-1600/.

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Arthur Pesah. Tailoring 3d topological codes and decoders to biased noise. Master of research dissertation, University College London, 2021. URL https://github.com/artix41/artix41.github.io/blob/b4f401e0c12b44be4c69318e0015f70d8a3d3f98/assets/pdf/master-thesis-ucl.pdf. Supervisor: Dan Browne. Submitted 1 September 2021.

Joschka Roffe, David R. White, Simon Burton, and Earl Campbell. Decoding across the quantum low-density parity-check code landscape. *Physical Review Research*, 2:043423, 2020. doi: 10.1103/PhysRevResearch.2.043423.

David K. Tuckett, Stephen D. Bartlett, Steven T. Flammia, and Benjamin J. Brown. Fault-tolerant thresholds for the surface code in excess of 5% under biased noise. *Physical Review Letters*, 124: 130501, 2020. doi: 10.1103/PhysRevLett.124.130501.

A Reproduction artifacts

The preserved runnable pure-X revision is 47f111b60fbf668edc8fa6dfb4fb6a398d6da933. It does not contain this expanded study. The immutable expanded runnable reproduction is expansion/READ

ME.md at d1a437af56a24b4d35f6511f6be0d84d86a7157c. New master navigation links await merge. The retained template is approved-preamble.tex. See coverage.json, expansion/README.md, threshold-comparisons.json, ranking-comparisons.json, expanded-manuscript.tex and expanded-references.bib. Main paper: biased_noise_thresholds_of_cubic_and_rhombic_codes__replication_of_pesah_2021.pdf.

B General algebra: proof-to-source correspondence

The earlier immutable proof/diagnostic milestone is cubic-proof-bridge.md at c3fbf87fa1ee4d414554cfa689fbc8aa065cca9f. Earlier milestone revisions below remain preserved. The immutable follow-up runnable revision is followup/README.md at 8fac693b11097f44406ea699e2e502ce71461bb1. It contains raw trials, frozen plans, analysis, validation and Lean source. Build with Lean 4.22.0 and Std only; build.sh records the commands. All reported theorems compile without admitted proofs or added axioms; earlier printed dependencies are `propext` and `Quot.sound`; the new existence/classification theorems additionally use standard `Classical.choice`. No new axioms are introduced. The following are standard algebraic reconstructions, not novelty claims.

Source obligation	Exact checked statement	Remaining bridge
Hadamard, 2.3.3/3.2	`binary_parameters_preserved`: arbitrary finite qubits and swap mask preserve isotropic subgroup, basis-witness rank and minimum logical weight.	Binary Pauli modulo-phase semantics; Hilbert-space dimension/phase admissibility and actual lattice parameter formulas are separate.
Cubic chains, 3.2.1	`chain_kernel`: periodic first-axis differences vanish iff constant along each chain.	Actual six-edge vertex formula now connected in Lean; cross-language finite matrix check added. Written counting and ideal-majority limit are in Section 7.
Cubic planes, 3.4	`plane_kernel`: periodic y/z differences vanish iff constant on each yz-plane.	Actual four-edge face formulas now connected in Lean. Written counting/probability in Section 7; no BP threshold implication.
Rhombic triangles, 3.5.2	`actual_two_corner_family_injective`: two restricted corner orientations expose every first-axis coordinate.	Formula-to-code translation manually audited; full even-periodic checkerboard geometry and Pauli convention remain separate.

A binary Pauli is a function $\{0,\dots,n-1\} \to \mathrm{F}_2^2$. The deformation swaps its components on a chosen mask. The proof verifies involution, XOR linearity, support and full-qubit symplectic-product preservation. It then transports subgroup membership, normalizers, logical cosets, a generating independent list and exact minimum-weight witnesses. The definition of encoded qubits is the explicit basis-rank relation $n = r + k$, not an independently proved Hilbert-space dimension theorem. `first_axis_parameters` instantiates $n = 3L^3$ and the mask $i < L^3$, matching the Python first-third swap. No CSS sector-distance invariance is asserted after mixed deformation.

The periodic kernel proof uses induction around one coordinate: equality of adjacent bits implies constancy; applying it twice gives constant planes. In the cubic implementation, vertex restriction supplies

neighboring first-axis edges (with a cyclic index shift), while the two face orientations supply y/z differences. The formal difference model allows positive lengths; the actual source lattice is used only at its stated admissible sizes, not a degenerate length-one implementation.

For rhombic triangles, the source reconstruction has one edge per axis and first-axis inward bit equal to base-X XOR vertex parity. Under first-axis-only error the other terms vanish. Orientations 0 and 2 supply base-X bits 0 and 1. Choosing base-X equal to vertex parity exposes the unshifted edge at every vertex, proving restricted-formula injectivity. This proof is stronger than the separate trivial coordinate-observation helper, but still does not formally extract the Python incidence matrix. The full code requires even periodic sizes; no odd-size physical result is claimed.

New source-linked statements in `CubicIncidence.lean` define all six vertex terms and all three four-edge face orientations on periodic coordinate triples. `cubic_incidence_commutes` proves that faces annihilate arbitrary vertex gradients: each corner bit occurs twice. `vertex_gradient_kernel` proves that zero gradient is equivalent to global constancy. The actual restricted chain and plane kernels are now connected to these formulas, including the periodic backward/forward inverse. `chain_syndrome_classification` characterizes equal restricted vertex syndromes by chain-constant XOR differences; component parameters are unique. These are general formula proofs, not numerical extrapolations.

A separate Lean consumer exports incidence entries, which are compared with the Python construction: six matrices at $L = 2, 4, 6$, 609,792 entries and zero mismatches. Implementations share geometric conventions but not the matrix generator. This is an exact finite cross-language bridge; arbitrary-size array flattening remains manually audited. The abstract XOR formulas allow positive lengths, whereas concrete matrix claims exclude degenerate length one. No general rank-number, logical-dimension or distance claim follows without its additional counting/homology proof.

At that earlier milestone, the written quantum, rhombic, majority and OSD bridges remained open. Section 7 assesses them; optional formalization of inherited background is not required for replication. The two rejected local fitting models are not repeatedly refitted. Historical production and GUI blockers remain distinct. After an initial ownership-claim denial, explicit user permission authorized one retry. Creation and remote readback confirmed the public claim; the coordination prerequisite is now met. Its history is recorded in `coordination.json`.

C Constructive periodic classification and distance

The immutable runnable continuation is `cubic-homology.md` at `d75bfba81d625f2d551e2b1df88e06d4eec63a1e`. Its `CubicPotential.lean`, `CubicDistance.lean` and `CubicLift.lean` contain the exact statements; the same pinned build script compiles them. Axiom reports contain only the declared standard axioms, including `Classical.choice` for existence/classification. These are reconstructions of the source's standard cubic-code claims, not mathematical novelty claims.

For a closed binary edge field on the unwrapped lattice, integrate along x at $y = z = 0$, then y at $z = 0$, then z. A proved XOR telescoping identity and the three plaquette equations show that the resulting potential has the original field as its gradient. This potential is not assumed periodic. Period shifts change it by position-independent bits, precisely the three cycle sums. A periodic potential exists if and only if those bits vanish. Two closed fields therefore differ by a periodic gradient if and only if their labels agree. Seam representatives realize every label; constant fields would fail to realize nonzero labels at even L. The finite adapter proves the analogous zero-holonomy criterion for the actual finite-coordinate face and vertex formulas, including the periodic boundary.

A nonzero axis cycle sum is the same on all L^2 parallel lines. Each must contain an occupied edge, giving a kernel-checked lower bound L^2. A seam representative attains it. This is the exact minimum

nontrivial cocycle weight in the coordinate model. The independently executable consumer uses actual Python vertex/face matrices and logical strings to check Lean-computed potentials, labels, weights and period jumps; all 80 constructed fields pass. General claims rely on proofs, not the number of examples.

For the written rank consequence, the constant vertex-gradient kernel gives rank L^3-1. The linear cycle-label map is onto three bits and its kernel consists exactly of gradients, so the closed-field dimension is L^3+2. Rank-nullity gives face rank $2L^3-2$, hence the CSS count $3L^3-(L^3-1)-(2L^3-2)=3$. Finite-dimensional rank-nullity/cardinality conversion, array-index rank correspondence and Hilbert-space interpretation are not all formalized; this is a written consequence of the checked structural results, not a Lean matrix-rank theorem.

For the written X-distance argument, finite binary annihilator duality pairs a non-boundary cycle with a nonzero seam class. Translated parallel seams differ by a vertex gradient, so a zero-syndrome cycle pairs identically with all L disjoint seams. Its weight is at least L; a straight periodic string attains L. This is a written analytical check, not a Lean duality theorem. The main analytical section supplies separate rhombic, probability, algorithmic and quantum arguments; no cubic result is transferred to the rhombic code without its own geometry.